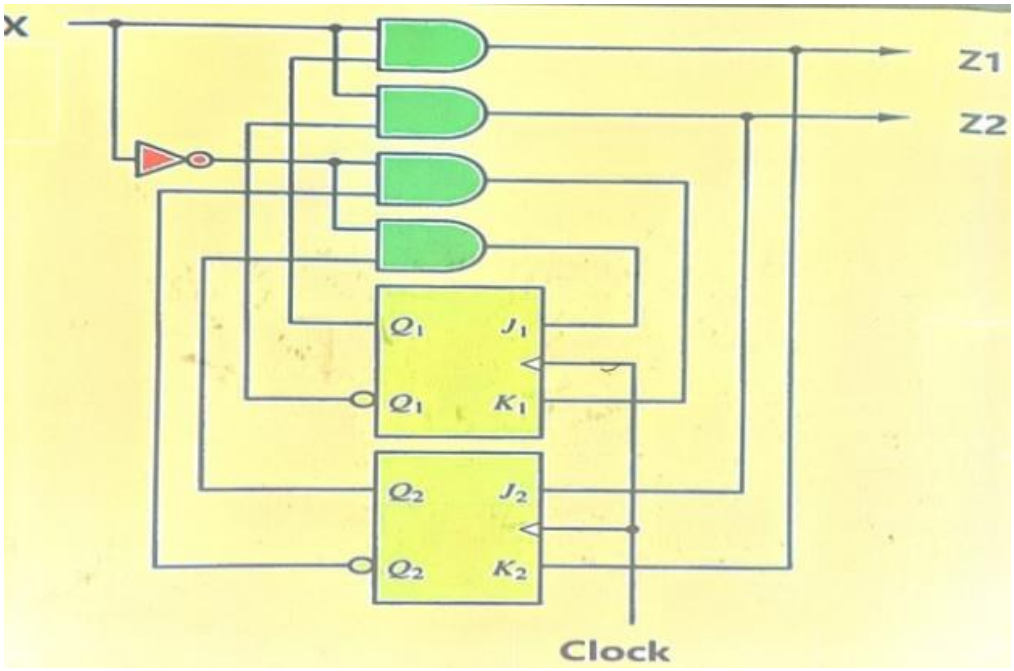
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Kelas: 2B-TNK

Rangkuman Minggu 13 Elektronika Digital Lanjut

1. Analisa rangkaian berikut dan buatkan state diagramnya!



Karena output Z di pengaruhi oleh input X maka rangkaian tersebut merupakan jenis Meally.

- Menetukan Persamaan :

Persamaan Output :

$Z_1 = XQ_1$

$Z_2 = X\overline{Q1}$

Persamaan Input :

$J_1 = Q_2\bar{X}$

$K_1 = \bar{X}\overline{Q2}$

$J_2 = Z_2 = X\overline{Q1}$

$K_2 = Z_1 = XQ_1$

- Kmap

Q_1Q_2\X	0	1
00	0	0
01	1	0
11	1	0
10	0	0

$J_1 = \bar{X}Q_2$

Q_1Q_2\X	0	1
00	1	0
01	0	0
11	0	0
10	1	0

$K_1 = \bar{X}\overline{Q2}$

Q_1Q_2\X	0	1
00	0	1
01	0	1
11	0	0
10	0	0

$J_2 = Z_2 = X\overline{Q1}$

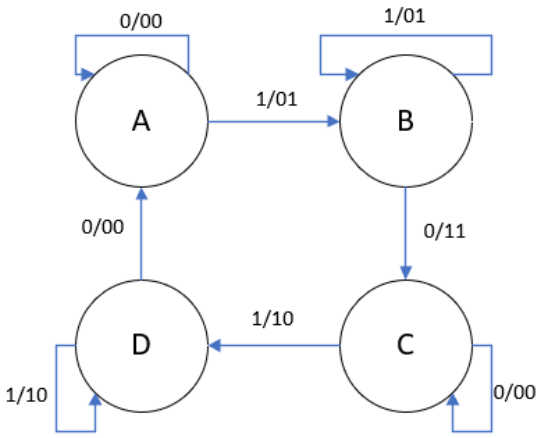
Q_1Q_2\X	0	1
00	0	0
01	0	0
11	0	1
10	0	1

$K_2 = Z_1 = XQ_1$

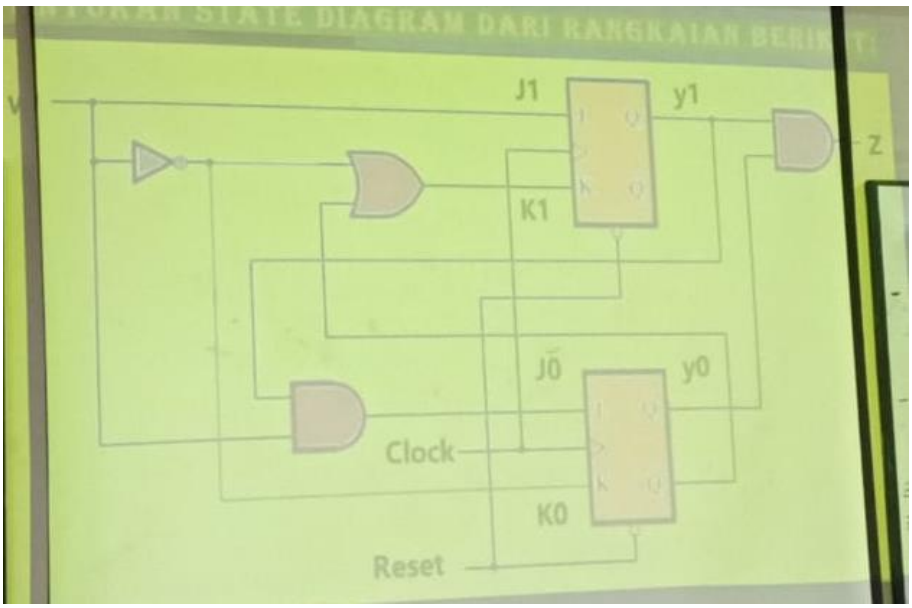
- State Table

PS	NS,Z_1Z_2	
	X = 0	X = 1
00	00,00	01,01
01	11,00	01,01
11	11,00	10,10
10	00,00	10,10

- State Diagram



2. Analisa rangkaian berikut dan buatkan state diagram nya!



- Menentukan Persamaan :

Persamaan Output :

$Z = Q_1Q_0$

Persamaan Input :

$J_1 = W$ $J_0 = Q_1W$

$K_1 = \bar{W} + \overline{Q0}$ $K_0 = \bar{W}$

- Kmap

Q_1Q_0\W	0	1
00	0	1
01	0	1
11	0	1
10	0	1

$J_1 = W$

Q_1Q_0\W	0	1
00	1	1
01	1	0
11	1	0
10	1	1

$K_1 = \bar{W} + \overline{Q0}$

Q_1Q_0\W	0	1
00	0	0
01	0	0
11	0	1
10	0	1

$J_0 = WQ_1$

Q_1Q_0\W	0	1
00	1	0
01	1	0
11	1	0
10	1	0

$K_0 = \bar{W}$

Q_1\Q_0	0	1
0	0	0
1	0	1

$Z = Q_1Q_0$

- State Table

PS	NS		OUT
	W = 0	W = 1	
00	00	10	0
01	00	11	0
11	00	11	1
10	00	01	0

- State Diagram

